MARITIME PORT CALLS ANALYSIS: 2022-2023

Management Report

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Data Source: UNCTADstat

Business Scenario & Objective

A maritime and logistics consulting company wants to better understand international port-call characteristics across economies and vessel segments in 2022 and 2023. Management requires an exploratory analysis that identifies notable operational patterns, differences, and outliers, as well as year-over-year changes between the two periods, that may warrant deeper investigation.

Acting as the Junior Data Analyst for this simulated case, I will use real-world maritime data to develop an interactive Tableau analysis and communicate the most relevant findings to management.

Data source: UNCTADstat (United Nations Conference on Trade and Development)

Primary dataset: Port Calls (2022–2023)

Data usage: Real-world public maritime data used within a simulated business-analysis scenario.

Data & Preparation

The project began with an initial examination of the raw UNCTAD Port Calls dataset to understand its structure, dimensions, measures, units, and missing values before creating visualizations.

The data contains multiple economies and ship categories, with measures describing different aspects of vessel activity. Particular attention was given to understanding the meaning and units of fields such as:

GT (Gross Tonnage) — a measure related to vessel size.

DWT (Deadweight Tonnage) — cargo-carrying capacity.

TEU (Twenty-foot Equivalent Unit) — container-carrying capacity.

Median Time in Port (Days) — the median duration of port calls.

During data cleaning and preparation:

- Column names were reviewed and standardized to make them clearer for analysis.

- Numerical measures were kept separate from the accompanying textual “Not available or not separately reported” fields rather than combining numeric and string information into one column.
- Missing values were examined rather than automatically replaced with zero, since a missing value does not necessarily represent zero activity or capacity.
- The 2022 dataset was cleaned using the same structure and logic as the 2023 dataset to ensure compatibility between the two years.
- The cleaned 2022 and 2023 datasets were then combined vertically in Tableau using a Union, resulting in a single dataset containing 402 records.
- Because the original tables did not contain a dedicated year variable, Tableau's automatically generated Table Name field was used to create a calculated Year dimension, allowing each record to be identified as either 2022 or 2023.
- Existing exploratory worksheets originally developed using the 2023 dataset were subsequently filtered to 2023 so that the addition of 2022 data would not alter the original analyses.
- This preparation resulted in a consistent dataset that supports both detailed 2023 exploratory analysis and direct 2022–2023 comparisons.

2023 Anaylsis & Findings

The 2023 analysis revealed substantial differences in port-call characteristics across ship categories. Dry bulk carriers consistently recorded comparatively long median times in port across economies, while vessel categories with much larger average vessel sizes, particularly LNG carriers, generally did not show correspondingly longer port stays. The scatter analysis therefore showed no clear relationship between average vessel size and median time in port. The capacity analysis also revealed substantial gaps between average and maximum container-carrying capacity, showing that the maximum vessel capacity observed within an economy can be considerably higher than its average.

2022–2023 Comparative Analysis

The year-over-year analysis was conducted to determine whether the patterns identified in the 2023 data were specific to that year or were also visible in 2022. Two indicators were selected for comparison: median time in port across ship categories and average container carrying capacity across economies.

The comparison of median time in port showed that differences between ship categories remained visible across both years. Dry bulk carriers continued to record comparatively long port stays, supporting the earlier 2023 observation that port duration appears to differ substantially by vessel category. At the same time, the magnitude of the year-over-year change varied across ship categories and economies. This indicates that the analysis should not be reduced to a single overall increase or decrease in port time; the result depends on both the selected economy and vessel segment.

A second comparison examined average container carrying capacity (TEU). Initial exploration showed higher average capacity in 2023 for many economies with observations available in both years. However, further examination identified exceptions, demonstrating that the increase was not universal across the dataset. The appropriate conclusion is therefore not that container capacity increased everywhere, but that a tendency toward higher average capacity was observed across many of the comparable economies.

The comparison also exposed an important data-availability limitation. Some economy-year combinations were not available in both datasets—for example, Japan was present in the 2023 data but not in the corresponding 2022 observations used in the analysis. Overall, the two-year comparison strengthens some of the patterns identified during the 2023 exploratory analysis while also demonstrating that changes are not uniform across economies or ship categories. The available data can establish where differences occur, but additional operational and contextual variables would be required to determine why those differences occurred.

Dashboards

Two interactive Tableau dashboards were developed to consolidate the analysis and allow management to explore the findings at different levels. The first dashboard focuses on 2023 vessel and port-call characteristics, combining port time, vessel size, and carrying-capacity measures. The second provides a 2022–2023 year-over-year comparison, allowing users to examine changes in median port time and average container carrying capacity by economy.

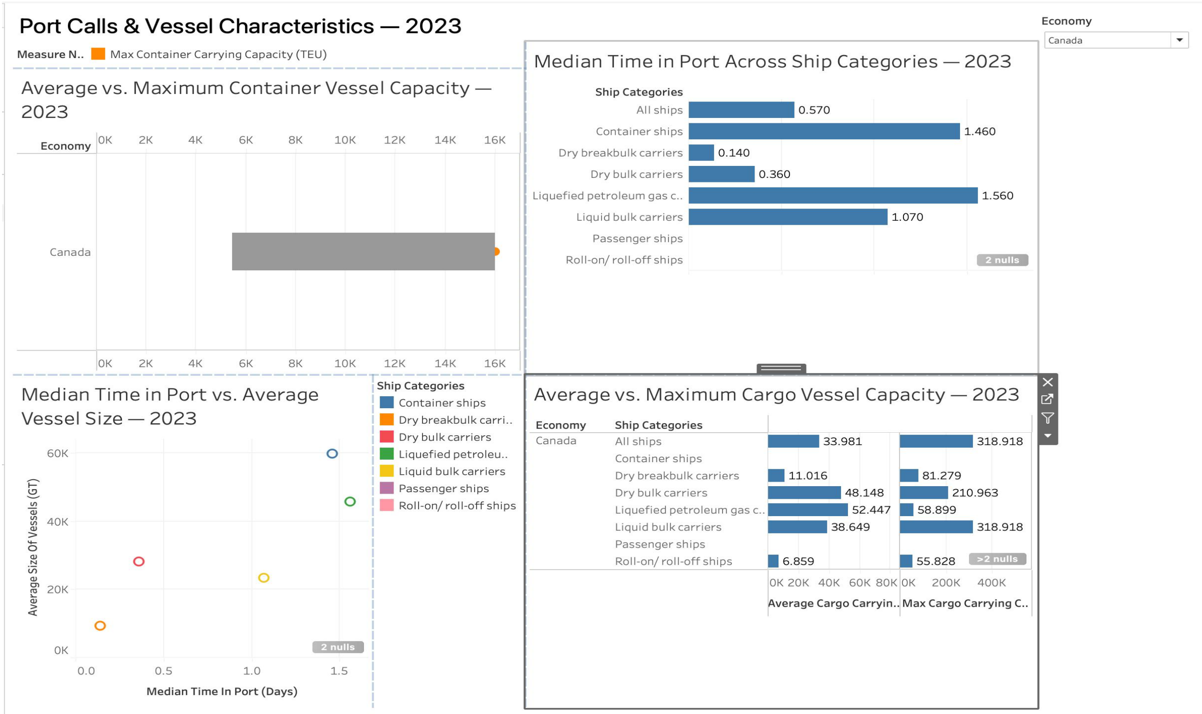

Figure 1. Port Calls & Vessel Characteristics - 2023

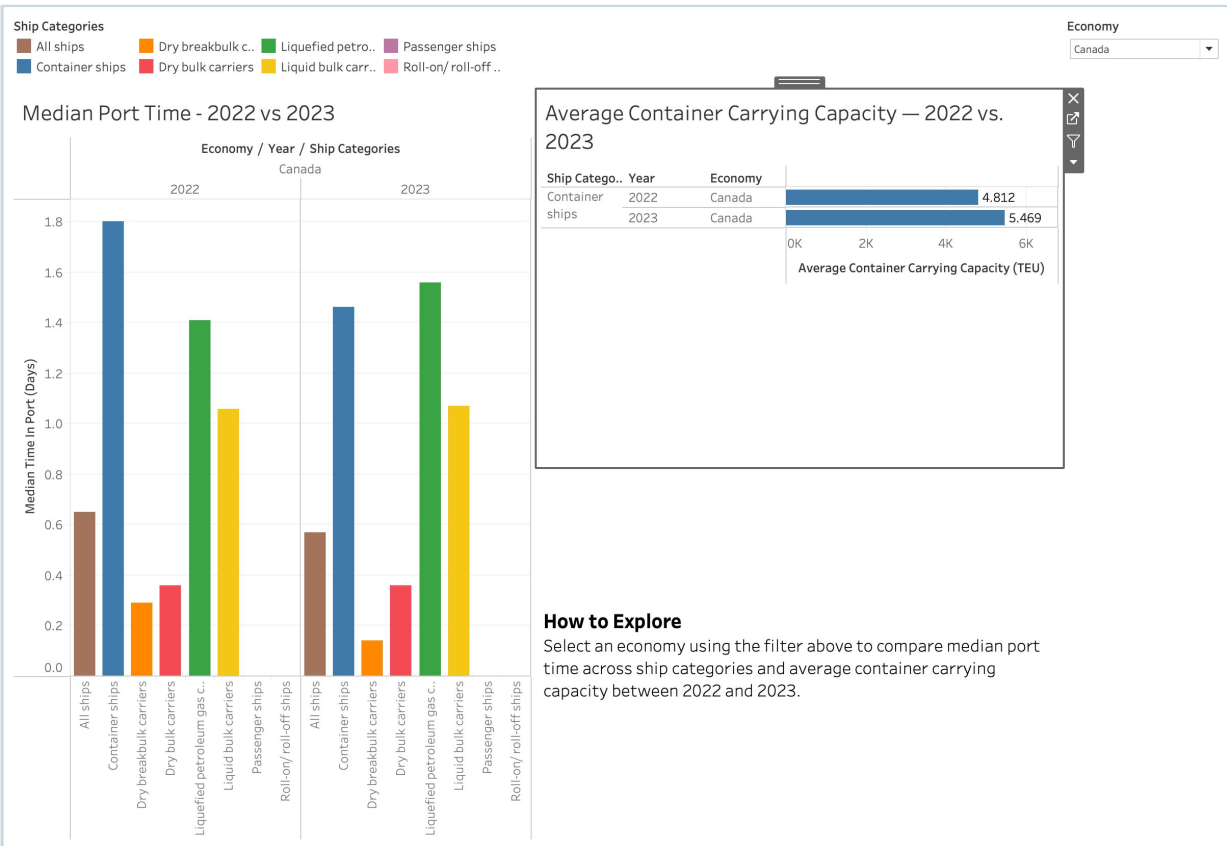

Figure 2. Year-over-Year Port Call Comparison — 2022 vs. 2023

MANAGEMENT TAKEAWAYS

The analysis highlights ship category as an important differentiating factor in port-stay duration, with dry bulk carriers consistently recording comparatively long median times in port across many economies and in both reporting years. At the same time, the analysis found no clear relationship between average vessel size and median port time, indicating that larger vessels do not necessarily correspond to longer port stays.

The year-over-year comparison also showed a tendency toward higher average container carrying capacity in 2023 across many comparable economies, although this pattern was not universal. Differences therefore need to be considered at the individual economy and vessel-segment level rather than interpreted as a uniform market-wide change.

From a management perspective, the consistently longer port stays observed for dry bulk carriers represent the clearest area for further investigation. However, the available dataset does not contain sufficient operational or contextual information to determine the underlying causes. Additional data would therefore be required before drawing conclusions about the drivers of these differences.